

Key Concepts

Entropy

Typical Set

Lossless Source
Coding

Channel Coding

Shannon
Capacity
Formula

ECE253/CSE208 Introduction to Information Theory

Lecture 15: Summary

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- *Elements of Information Theory (2nd Edition) by Thomas Cover & Joy Thomas*

Take-away Points

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- Entropy and differential entropy; KL divergence; mutual information.
- DPI; sufficient statistics; convex functions; Jensen's ineq; Fano's ineq.
- AEP; typical set; entropy rate; Markov chain; stationary stochastic process.
- Source coding theorem; Kraft-McMillan inequality; competitive optimality.
- Lossless source coding: Huffman, Shannon, SF, SFE, and arithmetic coding.
- Channel coding theorem and its proof; channel capacity of "simple" channels.
- Channel capacity of AWGN/parallel AWGN/feedback; water-filling power allocation.

For Discrete Distributions

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PMF $p(x)$	Entropy
Definition	$H(X) = -\sum_x p(x) \log p(x)$
Bounds	$[0, \log \mathcal{X}]$
$H_{\max}$ distribution	Uniform
Translation	$H(X + c) = H(X)$
Scaling	$H(cX) = H(X)$
Joint entropy	$H(X, Y) = -\mathbb{E}_{(X, Y)} \log p(X, Y)$
Conditional entropy	$H(Y X) = -\mathbb{E}_{(X, Y)} \log p(Y X)$
Relative entropy	$D(p q) = \sum p \log \frac{p}{q}$
Mutual information	$I(X; Y) = D(p(X, Y) p(X)p(Y))$
Chain rule	$H(X^n) = \sum_{i=1}^n H(X_i X^{i-1})$
AEP (i.i.d. X^n)	$-\frac{1}{n} \log p(X^n) \xrightarrow{\text{i.p.}} H(X)$
Typical set	$A_\epsilon^{(n)} = \{x^n : -\frac{1}{n} \log p(x^n) - H(X) \leq \epsilon\}$

For Continuous Distributions

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PDF $p(x)$	Differential Entropy
Definition	$h(X) = - \int_{\mathcal{S}} f(x) \log f(x) dx$
Bounds	$(-\infty, \frac{1}{2} \log(2\pi e \sigma^2)]$
$H_{\max}$ distribution	Gaussian
Translation	$h(X + c) = h(X)$
Scaling	$h(cX) = h(X) + \log c $
Joint entropy	$h(X, Y) = -\mathbb{E}_{(X,Y)} \log f(X, Y)$
Conditional entropy	$h(Y X) = -\mathbb{E}_{(X,Y)} \log f(Y X)$
Relative entropy	$D(f g) = \int f \log(\frac{f}{g})$
Mutual information	$I(X; Y) = D(f(x, y) f(x)f(y))$
Chain rule	$h(X^n) = \sum_{i=1}^n h(X_i X^{i-1})$
AEP (i.i.d. X^n)	$-\frac{1}{n} \log f(X^n) \xrightarrow{\text{i.p.}} h(X)$
Typical set	$A_{\epsilon}^{(n)} = \{x^n : -\frac{1}{n} \log f(x^n) - h(X) \leq \epsilon\}$

Typical Sequences and Set

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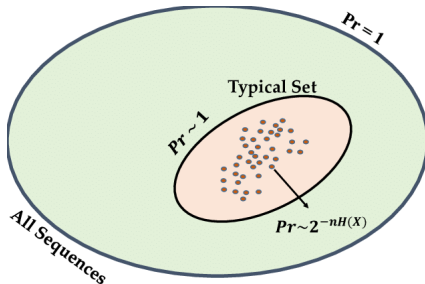


Figure: For n sufficiently large, all typical sequences have about the same probability $2^{-nH(X)}$ (*asymptotic equipartition*). Everything outside the typical set has a negligible probability.

Coding for Typical Set

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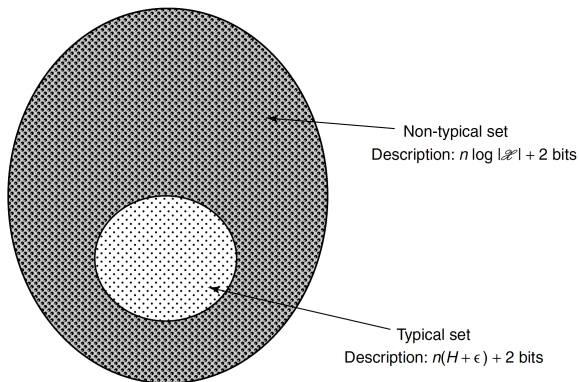


Figure: Encoding for the typical set: On average $H(X)$ bits are needed to encode X^n per symbol.

Lossless Source Coding (1/2)

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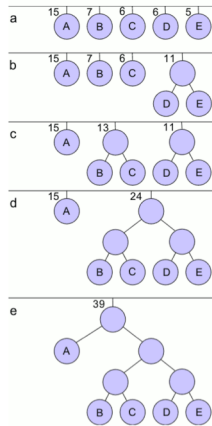


Fig 1. Huffman tree (bottom-up)

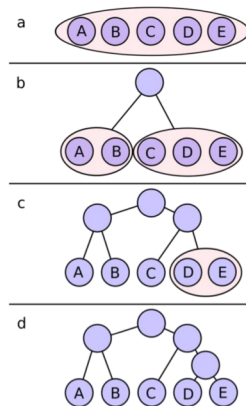


Fig 2. SF tree (top-down)

Lossless Source Coding (2/2)

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Shannon Capacity Formula

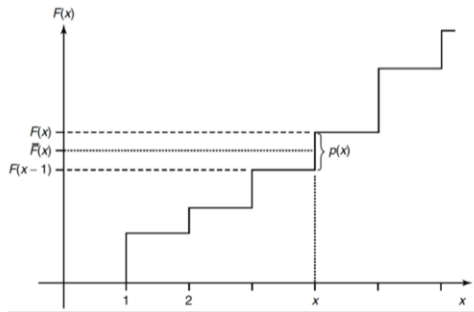


Fig 3. SFE coding

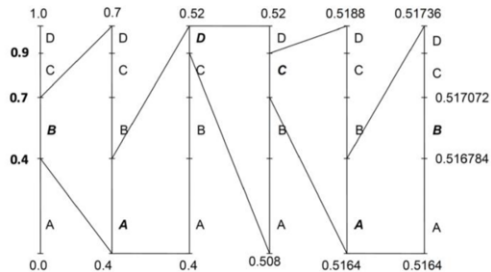


Fig 4. Arithmetic coding

Channel Coding Theorem

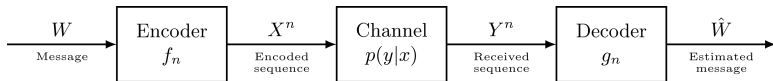


Figure: Channel capacity C is the sharp threshold between reliable and unreliable communication.

- *Information* channel capacity: $C = \max_{p(x)} I(X; Y)$
- *Operational* channel capacity: number of bits that are transmitted reliably (with an arbitrarily small probability of error) per channel usage.
- Information channel capacity = operational channel capacity.
- **Achievability:** All rates below capacity $R < C$ are achievable.
- **Converse:** $(2^{nR}, n)$ code with probability of error $\lambda^{(n)} \xrightarrow{n \rightarrow \infty} 0$ must have $R \leq C$.

Shannon Limit

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$$C_{\text{AWGN}} = W \log_2 \left(1 + \frac{P}{N_0 W} \right)$$

$$\max_{P_i \geq 0} \sum_{i=1}^N \frac{1}{2} \log \left(1 + \frac{P_i}{N_i} \right) \quad (1a)$$

$$\text{s.t.} \quad \sum_{i=1}^N P_i = P \quad (1b)$$

Water-filling Solution

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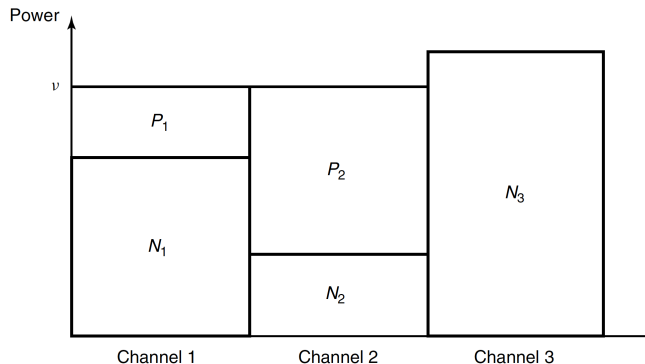


Figure: Water-filling optimal power allocation: **allocate more power in less noisy channels.**

By using the Lagrangian relaxation, the optimal solution is obtained as:

$P_i^* = (\nu - N_i)^+$, $i = 1, 2, \dots, k$, where the water level ν satisfies $\sum_i (\nu - N_i)^+ = P$.